

Review article

Queen lifespan and colony characteristics in ants and termites

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Summary

Comparative studies are increasingly used to identify factors responsible for the evolution of plant and animal phenotypes and social insects provide ideal systems for such studies as a result of their modular growth and their bewildering diversity in social organization. Here I review data on the lifespan of queens in ants and termites because such data are highly relevant for comparative analyses on life-history. Lifespan has been reported in 53 ant and 10 termite species. Information on the mode of colony founding and number of queens per colony is also provided for the 53 ant species. Finally, I suggest that students of social insects would greatly benefit if information on colony characteristics, individual phenotypes and behaviour would become available for a larger number of species, perhaps on a database that would be accessible on the web.

Introduction

Comparative studies are increasingly used to identify factors responsible for the evolution of plant and animal phenotypes (e.g., Harvey and Pagel, 1991; Leroi and Lauder, 1994). Social insects provide ideal systems for such studies as a result of their modular growth, their bewildering diversity in social organization, reproductive strategies, life-histories, diets, and their large number of species (Hölldobler and Wilson, 1990; Bourke and Franks, 1995; Crozier and Pamilo, 1996), yet the number of comparative studies that have been performed to date is very small compared to other taxa such as birds and mammals. There are perhaps no more than 20 comparative studies of social insects that include 10 or more species (e.g., Trivers and Hare, 1976; Cole, 1983; Keller and Passera, 1989; Passera and Keller, 1990; Frumhoff and Ward, 1992; Kaspari and Vargo, 1995) and very few of these studies have controlled for the effect of phylogeny (Frumhoff and Ward, 1992; Keller and Reeve, 1994; Keller and Genoud, 1997).

One important reason for the paucity of such comparative studies is the lack of detailed and reliable data on parameters such as colony size, queen number, worker and queen body sizes, number of matings per queen, and mating strategies. More-

over this information is frequently difficult to retrieve from the literature because it is often mentioned in a cryptic manner in papers dealing primarily with other issues. Moreover, valuable information on the composition of ant colonies is frequently never published because it does not fit with the general aim of the studies undertaken. Ideally, one would therefore like to have a database where students of social insects could deposit information collected during field work and other information that would be useful for comparative studies. However, the organization of such a database might be difficult as it probably would require considerable work to be organized. Review papers on colony characteristics and/or phenotypes and behaviour of colony members are therefore very useful.

The aim of the present paper is to make available information on the lifespan of ant and termite queens that I collected from the literature and unpublished observations by colleagues. The primary aim was to test evolutionary theories on ageing by investigating whether the evolution of eusociality had been paralleled by an increase in queen lifespan [the expected pattern if lifespan is influenced by extrinsic mortality (Alexander et al., 1991; Keller and Genoud, 1997)]. Unfortunately, space limitation prevented us from providing detailed information on queen lifespans and the relevant references. The publication of the full data set should be particularly useful because data for several species come from unpublished observations or papers published in French. As data on queen number (monogyne versus polygyne) and mode of colony founding were also collected for our analysis I also provide these data and the relevant references. These informations on queen lifespan and social organization are based on an extensive literature research and the personal observations of many colleagues.

Results and discussion

Information on lifespan was found for 54 ant species (Table 1) and 10 termite species (Table 2). The most striking pattern emerging from these Tables is the very long lifespan of queens, with values higher than 20 years in some species. An inspection of Tables 1 and 2 reveals that most data come from queens reared in the laboratory, typically with only few colonies studied. Hence, these lifespan estimates should be regarded as approximate and only reflect lifespan under laboratory conditions. For some species the maximum lifespan is underestimated because the queen died from the lack of water, escaped from the colony, or simply was still alive at the time of publication. Insufficient information was given for most of species to determine whether more than one colony had been followed in the laboratory and to assess the average lifespan of queens. Field data are also not without problems in most cases. For example, when the lifespan of queens is based on the lifespan of colonies it may be that some queens are replaced thus increasing colony longevity and the presumed queen lifespan. Queen replacement has indeed been recently shown to occur in monogyne colonies of the fire ant *Solenopsis invicta* (DeHeer and Tschinkel, 1998).

A previous analysis controlling for the phylogeny (Keller and Genoud, 1997) showed a significant association between queen lifespan, and queen number (monogyne versus polygyne $P < 0.001$) and the mode of colony founding ($P < 0.001$).

Table 1. Lifespan (years), queen number and mode of colony founding in ants. M: species that are primarily monogynous. P: species that are primarily polygynous. I: species that found new colonies primarily through independent colony founding; D: species that generally use a dependent colony founding; Pa: species that parasitize colonies of other ant species. Species that commonly have both monogyne and polygyne colonies are referred as M, P. Similarly, species that found new colonies independently and dependently are referred as I, D

Species	Max. lifespan	Mean lifespan	Ref.	Queen number	Ref.	Colony founding	Ref.	Comments
<i>Aphaenogaster rudis</i> ^c	13.1	8.7	13	M	15, 21	I	64	based on 11 queens in laboratory nests
<i>Atta cephalotes</i>	13.0 ^h		14	M	61, 72	I	61, 73	based on 1 queen in laboratory nest
<i>Atta sexdens</i>	15.3 ⁱ		20	M	61, 73	I	59, 61, 72	based on 1 queen in laboratory nest
<i>Camponotus consobrinus</i>	> 7		17	M	38	I	59, 38	based on 1 queen in laboratory nest
<i>Camponotus herculeaneus</i>	> 10		17	M	18	I	18, 19	based on 1 queen in laboratory nest
<i>Camponotus lateralis</i>	8.0 ^f		27	M	60	I	8	based on 1 queen in laboratory nest
<i>Camponotus perthiana</i>	23.2		4	M	5	I	4	based on 1 queen in laboratory nest
<i>Cardiocondyla emeryi</i> ¹	0.5	0.3	78	P	79	D	78	based on 38 queens in laboratory nests
<i>Crematogaster scutellaris</i>	20.0		24	M	24	I	24	based on at least 3 queens in laboratory nests
<i>Diacamma cf. rugosum</i>	1.9	1.6	75	M	75, 76, 77	D	77	based on 5 mated workers (= queens) in laboratory nests ^j
<i>Ectatomma ruidum</i>	8.8		5	M	23	I	23	based on 1 queen in laboratory nest
<i>Formica exsecta</i> ^a		20.0	1	M	1	I	8	based on estimation of age of field colonies with allozymes
<i>Formica fusca</i>	16.0		56	M	57	I	8	based on 2 queens in laboratory nests
<i>Formica polyctena</i>		5.0	68	P	68	D, Pa	8	based on survival of queens in field colonies
<i>Formica rufibarbis</i>	14.0		25	M	7	I	8	based on 1 queen in laboratory nest
<i>Formica sanguinea</i>	20.0		25	M, P	57	I, D, Pa ^g	22	based on 1 queen in laboratory nest
<i>Harpagoxenus sublaevis</i>	> 14	10.0	26, 35	M	26, 28, 35	Pa ^k	26, 35	based on several queens in laboratory nests and the age of field colonies
<i>Lasius alienus</i>	9.0		29	M	30, 31	I	8	based on 1 queen in laboratory nest
<i>Lasius flavus</i>	22.5		32	M	8, 30, 34	I	8	based on 1 queen in laboratory nest
<i>Lasius niger</i>	28.5		25	M	8, 30, 34	I	8	based on 1 queen in laboratory nest
<i>Leptothorax acervorum</i>		11.0	35	M, P	8	I	36	based on 2 queens in laboratory nests
<i>Leptothorax affinis</i>	12.0	11.0	36	M	36	I	36	based on many queens in laboratory nests

Table 1 (continued)

Species	Max. lifespan	Mean lifespan	Ref.	Queen number	Ref.	Colony founding	Ref.	Comments
<i>Leptothorax exilis</i>	>16		35	M	35	I	35	based on 1 queen in laboratory nest
<i>Leptothorax lichtensteini</i>	17.0	12–15	10	M	10, 36	I	10	based on many queens in laboratory nests
<i>Leptothorax melas</i>	>10	9.3	36	M	36	I	36	based on 5 queens in laboratory nests
<i>Leptothorax nylanderii</i>	19.0	15.0	9, 10	M	10, 36	I	10	based on many queens in laboratory nests
<i>Leptothorax parvulus</i>	12.0	10.0	10	M	10, 36	I	10	based on many queens in laboratory nests
<i>Leptothorax rabaudi</i>	>10	5.9	36	M	36	I	36	based on 15 queens in laboratory nests
<i>Leptothorax unifasciatus</i>	12.0	11.0	36	M	36	I	36	based on many queens in laboratory nests
<i>Linepithema humile</i>		0.9	43	P	43, 44, 45	D	46, 19	based on the observation that 90% of the queens are executed when they are about 10 month old in the field
<i>Messor semirufus</i>	9.0		37	M	39	I	37	based on several queens in laboratory nests
<i>Monomorium pharaonis</i>	0.7	0.5	47	P	48	D	48	based on 10 queens in laboratory nests
<i>Myrmecocystus mimicus</i>	>11		17	M	41	I	41	based on 1 queen in laboratory nest
<i>Myrmecia gulosa</i>	12.8		5	M	5	I	5	based on 1 queen in laboratory nest
<i>Myrmecia nigriceps</i>	7.2 ^d	5.8 ^d	5	M	5	I	5	based on 4 queens in laboratory nests
<i>Myrmecia vindex</i>	10.8	7.6	5	M	5	I	5	based on 11 queens in laboratory nests
<i>Myrmecia limanica</i>		1.8	50	M, P	50	I, D	12	Inferred from the rapid turn-over of queens in the field
<i>Myrmecia lobicornis</i>		< 2	11	M, P	11	I, D ^b	11, 12	Inferred from genetic data on colony composition in the field
<i>Myrmecia ruginodis</i>		< 1.5	11	M, P	11, 52	I, D ^b	11, 12	Inferred from genetic data on colony composition in the field
<i>Myrmecia sabuleti</i>	4.0	2.5	51	M, P	51, 52	I, D	12	Inferred from queen turnover
<i>Myrmecia sulcinodis</i>		4–5	53	M, P	52, 53	I, D	12	Inferred from the fluctuation of queen number per colony
<i>Myrmecia tahoensis</i>		< 2	70	M, P	70	D	70	Inferred from genetic data on colony composition in the field

Table 1 (continued)

Species	Max. lifespan	Mean lifespan	Ref.	Queen number	Ref.	Colony founding	Ref.	Comments
<i>Odontomachus haematoda</i>	4.5		5	M	5, 62	I	5, 62	based on 1 queen in laboratory nest
<i>Odontomachus insularis</i>	4.5		5	M	5	I	5	based on 1 queen in laboratory nest
<i>Pogonomyrmex badius</i>	18.0		16	M	30	I	55	based on 1 queen in laboratory nest
<i>Pogonomyrmex barbatus</i>	20.0	17.5	65	M	65, 66	I	65	Inferred from survival of field colonies
<i>Pogonomyrmex owyheei</i>	30.0	17.0	2	M	2	I	55	Inferred from survival of field colonies
<i>Rhytidoponera purpurea</i>	12.5		5	M	5, 33	I	5, 6	based on 1 queen in laboratory nest
<i>Sericomyrmex urichi</i>	11.0		49	M	40	I	49	based on 1 queen in laboratory nest
<i>Solenopsis invicta</i>		5.8–6.8	3	M, P ^c	71	I, D	71, 74	Inferred from the age of colonies and rate of sperm use of queens
<i>Stenamma westwoodi</i>	17–18		42	M	58	I	5	based on 1 queen in laboratory nest
<i>Tapinoma sessile</i>	a few weeks	a few weeks	69	P	69	D	69	based on many queens in laboratory nests
<i>Tetraponera anthracina</i>	>3.1		54	P	54, 67	D	67	based on 18 queens in laboratory nests
<i>Wasmannia auropunctata</i>		1.1	63	P	63	D	63	based on 12 queens in laboratory nests

^a The studied population was monogynous (Pamilo, 1991). There are no data for polygynous populations.

^b Genetic data (Seppa, 1994) show that queens are recruited in their own nest.

^c This species was originally referred as *Aphaenogaster picea* (B. Hölldobler, pers. comm.).

^d Lifespan is that given for queens plus one year (the age at collection).

^e The studied population was monogynous. However, native populations of this species usually contain both monogyne and polygyne colonies (Ross et al., 1996).

^f Palma-Valli (1981) also reports a minimum lifespan of 5.6 years for field colonies.

^g Colony founding occurs by several means (reviewed in Forel, 1922). Founding queens may, 1) steal worker pupae from other species (e.g., *Formica fusca* and *Formica rufibarbis*) to help to rear their own brood, 2) associate with founding queens of another species (e.g., *Formica fusca*) and 3) enter established colonies, leading to colony reproduction by budding.

^h Weber (1976a) also reports a lifespan of 10 years for a queen kept in a laboratory nest.

ⁱ K.M. Horton and E.O. Wilson (in Hölldobler and Wilson, 1990) also report a lifespan of 14 years for a queen kept in a laboratory nest.

^j There are no morphological castes in this species. Reproduction is monopolized by a single mated worker, and it is the lifespan of these individuals that is reported (Fukumoto et al., 1989; Peeters and Higashi, 1989; Tsuji et al., 1996).

Table 1 (continued)

^k This species was classified as independent founding by Keller and Genoud (1997) because queens kill all adult individuals of the host colonies thus preventing queens to get the assistance of workers from an established colony. However, the presence of the host brood gives a head start to colony development (J. Heinze, pers. comm.) so that this species would probably be best classified as intermediate between an independent and dependent founding species in terms of age-dependent increase in reproductive success (the important parameter that affects lifespan).

^l This species was not included in Keller and Genoud (1997)'s study because we were unaware of the data. However, this polygyne, dependent founding, species has a very low lifespan, as predicted by theory.

1 (Pamilo, 1991); 2 (Porter and Jorgensen, 1988); 3 (Tschinkel, 1987); 4 (Haskins and Haskins, 1992); 5 (Haskins and Haskins, 1980); 6 (Ward, 1981); 7 (Gösswald, 1989); 8 (Kutter, 1977); 9 (Plateaux, 1980); 10 (Plateaux, 1986); 11 (Seppa, 1994); 12 G. Elmes, pers. comm.; 13 (Haskins, 1960); 14 D. Cherix, pers. comm.; 15 (Headley, 1949); 16 K.M. Horton and E.O. Wilson in (Hölldobler and Wilson, 1990); 17 B. Hölldobler in (Hölldobler and Wilson, 1990); 18 (Hölldobler, 1962); 19 (Keller and Passera, 1989); 20 (Autuori, 1950); 21 (Crozier, 1973); 22 (Forel, 1922); 23 (Corbara et al., 1989); 24 (Casevitz-Weulersse, 1991); 25 H. Appel in (Kutter and Stumper, 1969); 26 (Buschinger, 1974); 27 K. Hölldobler in (Hölldobler and Wilson, 1990); 28 (Bourke et al., 1988); 29 (Janet, 1904); 30 (S. Cover in (Keller and Reeve, 1994); 31 A. Buschinger in (Frumhoff and Ward, 1992); 32 (Prescott, 1973); 33 (Ward, 1983); 34 (DumPERT, 1978); 35 A. Buschinger, pers. comm.; 36 L. Plateaux, pers. comm.; 37 (Tohme and Tohme, 1978); 38 B. Hölldobler, pers. comm.; 39 S. Cover, pers. obs.; 40 (Nonacs, 1986); 41 (Bartz and Hölldobler, 1982); 42 (Donisthorpe, 1936); 43 (Keller et al., 1989); 44 (Markin, 1970); 45 (Keller, 1988); 46 (Newell and Barber, 1913); 47 (Peacock and Baxter, 1950); 48 (Peacock et al., 1955); 49 (Weber, 1976b); 50 (Elmes and Petal, 1990); 51 (Brian, 1972); 52 (Elmes and Keller, 1993); 53 (Elmes, 1987); 54 (Terron, 1977); 55 S. Porter, pers. comm.; 56 (Lubbock, 1888); 57 (Rosengren et al., 1993); 58 (Bernard, 1968); 59 (Autuori, 1956); 60 (Provost, 1985); 61 (Wilson, 1985); 62 (Haskins and Enzmann, 1938); 63 (Ulloa-Chacón and Cherix, 1989); 64 (Haskins and Enzmann, 1945); 65 (Gordon, 1991); 66 D. Gordon, pers. comm.; 67 (Terron, 1970); 68 (Horstmann, 1983); 69 (Harada, 1990); 70 (Evans, 1996); 71 (Porter, 1992; Ross and Keller, 1995; Ross et al., 1996); 72 (Weber, 1972); 73 E.J. Fjerdingstad, pers. comm.; 74 (Tschinkel, 1996); 75 (Tsuji et al., 1996); 76 (Peeters and Higashi, 1989); 77 (Fukumoto et al., 1989) 78 J. Heinze, pers. comm.; 79 (Heinze and Trenkle, 1997).

Table 2. Lifespan (years) of queens in termites

	Max. lifespan	Mean lifespan	Ref.	Comments
<i>Cyclotermes</i> sp.	12		1	no indication available on the source of information
<i>Kalotermes minor</i>	14		2	based on several queens in laboratory nests
<i>Macrotermes bellicosus</i>	> 11	> 11	3	based on several queens in laboratory nests; several queens still alive
<i>Macrotermes michaelseni</i>	19	ca. 18	3	based on 8 queens in laboratory nests; several queens still alive
<i>Macrotermes subhyalinus</i>	> 17	ca. 16	3	based on 7 queens in laboratory nests; several queens still alive
<i>Mastotermes darwiniensis</i>		> 7	4	based on 1 queen in laboratory nest ^a
<i>Reticulitermes hesperus</i>	30		5	no indication available on the source of information
<i>Reticulitermes santonensis</i>	> 7 ^b	> 7	6	based on several queens in laboratory nests
<i>Zootermopsis angusticollis</i> ^d	5.9 ^c		7	based on 1 queen in laboratory nest

^a The queen died when she was between 7–17 years old. The male lived more than 17 years.
^b Buchli (1958) believes that queens can live 10–15 years, but he provides no data to support this claim.
^c Heath (1907) also found that most colonies lived less than 4.5 years in the field. In two colonies the royal pair was still alive after 4.5 years.
^d This species was originally (Heath, 1907) referred as *Termopsis angusticollis* (Grassé, 1984).

1 Beeson 1941 (in Wilson, 1971, page 431); 2 (Harvey, 1934); 3 R. Leuthold, pers. comm.; 4 (Watson and Abbey, 1989); 5 Pickens 1934 (in Grassé, 1984, page 146); 6 (Buchli, 1958); 7 (Heath, 1907).

The association between lifespan and queen number was expected because the extrinsic rate of mortality is predicted to be higher in polygyne than in monogyne species (Nonacs, 1988; Keller and Genoud, 1997). Queen lifespan was also associated with the mode of colony founding, with independent founding queens (i.e., those starting new colonies without the help of workers) living longer than dependent founding queens (i.e., those unable to start a new colony without the help of workers). This pattern was also expected by evolutionary theories of ageing because the reproductive success of independent founding queens increases more with age than that of dependent founding species and an age-dependent increase in reproductive success should select for greater lifespan (Stearns, 1992). The higher age-dependent increase in reproductive success of independent founding queens stems from the fact that it usually takes several years until an independent founding queen

can produce sexuals, whereas queens returning to established nests after mating may produce sexuals sooner (Keller and Passera, 1990; Keller, 1993).

Inspection of Table 1 reveals tremendous variation in lifespan among species having the same type of social organization. These differences are likely to be associated with specific differences in extrinsic mortality and differences in age-dependent increase in reproductive success. It would be interesting to investigate how queen lifespan correlates with other factors such as colony size and queen/worker polymorphism. Unfortunately, more data on queen lifespan, colony sizes, queen/worker polymorphism and other relevant parameters are needed before such analyses are worth performing.

Oster and Wilson (1978) also made interesting predictions about differences in lifespan among members of different worker castes. They suggested that minor workers should in general have the shorter survivorship schedule, since when older they forage almost daily, whereas major workers leave the nest less frequently, thus experiencing lower extrinsic mortality. They also suggested that average lifespan should be the longest in species such as *Colobopsis Zacryptocerus* where the major workers almost never go outside the nest. As far as I know such tests have never been conducted. While doing such tests it would be important to consider alternative explanations such as higher lifespan of the larger worker caste being due to their typically lower metabolic rate. Ideally one would thus like to compare lifespan in species where smaller individuals experience lower extrinsic mortality than individuals of the largest caste. In such a situation evolutionary theories and purely mechanistic theories of ageing would make the opposite prediction on the association between caste membership and lifespan.

I encountered several problems while classifying species as monogyne or polygyne. In most cases species were categorized without any indication of the number of colonies surveyed or the number of queens found per nest. It is becoming increasingly clear that species that always have only one queen per colony and species that always have multiple queens per colony are only endpoints on a monogyny-polygyny continuum. Most species probably form both single and multiple queen colonies. Hence, species should better be classified on a monogyny-polygyny continuum with either the proportion of polygyne colonies or the average queen number per colony as a measure of the "level" of polygyny. Unfortunately, such information is available for only a minority of the species. An additional problem is that the number of queens per colony may vary among seasons, as for example has been reported in the Argentine ant (Keller et al., 1989) and species of the genus *Leptothorax* (Herbers and Stuart, 1996; J. Heinze, pers. comm.).

Similar problems were encountered with the mode of colony founding. A significant number of species may adopt several alternative strategies as exemplified by many species of the genus *Formica* where young queens can either stay in their mother nest after mating, enter the nest of a conspecific colony or parasitize other ant species (Cherix et al., 1991). However, in most cases it is unknown which of the alternative strategies is most common.

To conclude, students of social insects would greatly benefit if information on colony characteristics, individual phenotypes and behaviour would become available for a larger number of species. Tschinkel (1991) has already advocated for more studies on what he called the "sociometry" of insect colonies. Such studies will

be particularly valuable if information is given on the number of individuals, colonies, and/or populations from which the data were collected. Finally, if any colleague would have the capacity and time to set up a database where researchers could deposit information on colony characteristics and other variables that would be useful for comparative studies, this would certainly be of great help to our field in general.

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